

Enterprise Intelligence & Investment Research Platform

Complete Feature Requirements, Developer Implementation Plan, Data Architecture, Scaling Strategy, Cost Optimization, and Claude Financial Skills Integration

Version 1.0 - Developer Handoff Specification

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Purpose

This document translates the product concept into a buildable enterprise software specification. It includes feature requirements, data architecture, source connectors, cost-aware scaling strategies, API-vs-ingestion alternatives, implementation workstreams, Codex development workflow, and explicit integration of Claude's newly released financial-services agents and skills for investment research, valuation, earnings, due diligence, and report generation.

Core Promise

Given a company, ticker, investor, fund, government agency, bill, regulation, contract, or person, the platform builds an evidence-backed intelligence graph, analyzes financial and market data, identifies related parties, and produces detailed reports that analysts can edit, enhance, review, approve, and export.

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1. Executive Summary and Recommended Architecture

The platform should be built as a multi-tenant, evidence-first intelligence system. The application should not place every raw record directly inside a warehouse or graph database. Instead, it should separate raw evidence, normalized operational data, analytical fact tables, search indexes, vector indexes, and graph views. This separation minimizes long-term storage and compute cost while making data lineage and report citations reliable.

The recommended architecture is a hybrid lakehouse design: Postgres for operational entities and permissions; object storage for raw evidence; Apache Parquet plus Apache Iceberg or Delta Lake for the lakehouse; ClickHouse or BigQuery/Snowflake for high-volume analytics depending on budget and enterprise customer expectations; OpenSearch for document and semantic retrieval; a graph layer starting in Postgres and moving to Neo4j or another graph database when pathfinding workloads justify it.

Recommended default stack

- Application and operational state: Postgres with partitioned tables, read replicas, and pgvector only for smaller or high-value vector workloads.
- Raw evidence and source documents: S3-compatible object storage using immutable paths, document hashes, retention policies, and lifecycle tiers.
- Lakehouse: Parquet files partitioned by source/date/entity, with Iceberg as the preferred open table format for vendor-neutral large analytical tables; Delta Lake is acceptable if the team chooses a Databricks-heavy stack.
- Analytics warehouse: start with ClickHouse or serverless query engines for cost-efficient event analytics; add Snowflake or BigQuery for enterprise customers who require managed warehouse governance and BI integrations.
- Search: OpenSearch for full-text, hybrid search, and vector search at production scale; avoid embedding every document until the value is proven.
- Graph: start with relational edge tables; graduate selected graph workloads to Neo4j or another graph database once relationship traversal, pathfinding, and graph algorithms become bottlenecks.
- Orchestration: Dagster for asset-lineage-heavy pipelines, or Prefect for faster Python-native workflow development. Either is acceptable, but choose one for version 1.

1.1 Key architectural decision

Do not build this platform as a single warehouse, a single graph database, or a single LLM/RAG application. It should be a system of specialized stores connected by strong evidence IDs. The lowest-cost long-term architecture is: raw objects in a lake, normalized facts in Postgres, analytical tables in a lakehouse/warehouse, searchable documents in OpenSearch, optional graph projections for pathfinding, and report claims backed by immutable evidence references.

1.2 Enterprise value proposition

- For investors: analyze public companies, stocks, financial statements, catalysts, valuation, technicals, government exposure, legal risk, lobbying, contracts, and related parties.
- For analysts: build evidence-backed reports from SEC, FEC, LDA, FARA, Congress, Federal Register, Regulations.gov, USAspending, SAM.gov, IRS 990, CourtListener/RECAP, sanctions, OpenCorporates, GLEIF, market data, earnings, and licensed research sources.
- For enterprises: collaborate across workspaces, enforce permissions, review claims, audit user activity, and export reports, decks, and models.
- For developers: use a standard connector contract, evidence model, entity graph schema, and report pipeline to add sources and analysis modules safely.

1.3 Developer success criteria

- Every normalized entity and edge has at least one EvidenceRef.
- Every connector stores raw source records before parsing.

- Every report claim has source evidence, confidence, and review status.
- Every expensive workflow has a cost budget and caching strategy.
- Every user-facing conclusion is editable in review and then re-verified before export.
- Every financial model exposes assumptions, source data, and scenario sensitivity.

2. Product Scope and Enterprise Users

2.1 Product mission

Build an enterprise intelligence and investment research platform that connects public records, market data, filings, regulations, lobbying, procurement, legal records, sanctions, corporate registries, news, and licensed financial data into one evidence-backed research environment. The system should operate as an analyst team in software form: financial analyst, equity researcher, investigative researcher, public-records analyst, government-affairs analyst, procurement analyst, legal-risk analyst, and report writer.

2.2 Supported users

Role	Primary responsibilities	Key permissions
Organization owner	Manage tenant, billing, SSO, retention, source policies, global settings.	Full admin; manage org, users, workspaces, sources, audit logs.
Workspace admin	Manage team workspace, roles, cases, watchlists, report templates.	Workspace admin; configure sources, invite users, approve exports.
Senior analyst	Lead investigations, approve reports, review claims, handle entity merges.	Create and approve reports; approve high-risk claims; merge/unmerge entities.
Analyst	Research entities, run analyses, create draft reports, add evidence and notes.	Search, analyze, draft reports, propose merges, create watchlists.
Investor user	Analyze stocks, portfolios, catalysts, valuation, and risk.	Run stock analysis, valuation reports, portfolio reports, watchlists.
Compliance reviewer	Review sensitive claims, export risk, source restrictions, investment-language risk.	Approve/reject claims, review reports, enforce restricted data policies.
Guest reviewer	Review assigned drafts, comment, and approve sections if allowed.	Limited case/report access, comment-only or restricted edit.

2.3 Business domains

- Equity research and stock analysis: public-company fundamentals, valuation, technicals, earnings, ownership, catalysts, risk, and reports.
- Public-record intelligence: corporate records, SEC filings, lobbying, political money, government contracts, regulations, public comments, court records, nonprofits, sanctions, and exclusions.
- Investor due diligence: related parties, financial quality, governance, legal risk, regulatory exposure, procurement concentration, and hidden dependency analysis.
- Enterprise collaboration: multi-user workspaces, audit trails, review workflows, report approvals, comments, versioning, and secure exports.
- Data platform: source connectors, evidence vault, data lake, warehouse, graph, search, vector retrieval, monitoring, and cost controls.

2.4 Product modes

Mode	User intent	Core outputs
Day trader	Find short-term catalysts, price action, liquidity, and risk levels.	Pre-market scans, volume spikes, technical setup, news/filing catalysts, risk/reward brief.
Swing trader	Identify 1-12 week price catalysts with confirmation from filings/news/regulatory events.	Catalyst calendar, trend state, volatility, trade thesis, invalidation levels.
Long-term investor	Understand business quality, valuation, durable catalysts, and multi-year risks.	Investment thesis, DCF, comps, risk map, management/governance review, monitoring plan.
Government exposure	Understand contracts, grants, agencies, lobbying, regulation, procurement concentration.	Agency map, lobbying paths, contracts, grants, public comments, OIRA/regulatory exposure.
Investigative research	Identify related parties, hidden links, public-record paths, and evidence-backed claims.	Graph paths, timeline, evidence appendix, confidence labels, related-party report.
Enterprise due diligence	Assess counterparty, vendor, portfolio company, executive, or fund risk.	KYB-style pack, sanctions/exclusions, litigation, ownership, procurement, regulatory risk.

3. Final Target Flow with Review, Editing, and Enhancement

The final target flow must support both automated generation and human-led refinement. Reports and proposals should not be static outputs. They should be living artifacts that can be edited, enhanced, re-run, commented on, reviewed, verified, and exported.

3.1 Full target user flow

- 1 User searches for a company, ticker, investor, person, fund, agency, bill, regulation, docket, contract, or portfolio.
- 2 The platform resolves the entity, shows aliases and identifiers, and asks the user to confirm the target entity if confidence is low.
- 3 The platform gathers evidence from operational stores, the data lake, search index, graph edges, market data, SEC data, public records, and licensed sources.
- 4 The user chooses a workflow: Analyze Stock, Company Intelligence Report, Investor Memo, Government Exposure Report, Related Parties Report, Diligence Pack, Proposal/Pitch, Portfolio Report, or Custom Investigation.
- 5 The workflow engine builds an Evidence Bundle and runs specialized analysis modules, including financial analysis, public-record analysis, graph pathfinding, and Claude financial skills where configured.
- 6 The draft report is generated with claim cards, source links, confidence labels, assumptions, tables, charts, valuation outputs, and open questions.
- 7 The draft enters Review Workspace. Users can edit text, comment, request enhanced sections, ask for alternative analysis, regenerate subsections, attach extra evidence, add charts, or run a Claude finance skill on selected sections.
- 8 Every edit triggers claim re-verification. If an edit changes a factual claim, the report engine requires evidence or marks the claim as unsupported.
- 9 A reviewer approves or rejects claims, comments, models, valuation assumptions, report sections, and final export.
- 10 Approved reports can be exported to PDF, Word, PowerPoint, Excel model, Markdown, HTML, JSON, and evidence appendix.
- 11 The entity or portfolio can be added to watchlists so the system alerts users when new filings, contracts, lobbying reports, regulations, court records, or stock catalysts appear.

3.2 Review Workspace requirements

Capability	Requirement
Section editor	Rich text editor with section-level locking, comments, suggested edits, source sidebar, and tracked revision history.
Enhance with AI	User can select a section and request: improve clarity, make investor-grade, add risk factors, expand valuation analysis, add counterargument, convert to executive summary, build a client proposal, or generate slide bullets.
Evidence sidebar	Every paragraph can show linked claims and EvidenceRefs. Analysts can drag evidence into a section to support a claim.
Claim cards	Each factual or analytical assertion is stored as a Claim object with confidence, sources, contradictions, and review status.
Citation verifier	After edits or AI enhancement, all claims are checked again. Unsupported claims are blocked or marked for review.
Version history	Each report and section has version snapshots, diff view, restore, reviewer comments, and export history.
Proposal mode	Reports can be converted into proposal/deck structures: client problem, research approach, findings, impact, timeline, deliverables, cost, next steps.
Financial model linkage	Excel models and valuation tables remain linked to report assumptions; modified assumptions trigger model recalculation and report-update suggestions.
Approval workflow	Draft -> analyst review -> compliance review -> final approval -> export/publish. High-risk claims require explicit approval.

3.3 Required edit and enhancement actions

- Rewrite section for executive audience.
- Expand section with more evidence from selected sources.
- Add bull/base/bear case.
- Add counterargument or short thesis.

- Add valuation assumptions and sensitivity table.
- Convert report into investor memo.
- Convert report into PowerPoint outline.
- Convert report into proposal for a client or internal committee.
- Refresh with latest filings and market data.
- Run Claude comparable-company analysis on selected peer set.
- Run Claude DCF model on selected financials and assumptions.
- Run Claude earnings-analysis skill on transcript and filing pack.
- Run Claude initiating-coverage skill for a full report draft.
- Verify all claims after changes.

3.4 Report states

State	Meaning	Allowed actions
Draft	Generated or manually created but not reviewed.	Edit, enhance, add evidence, rerun modules, delete, comment.
In review	Submitted to reviewer or compliance queue.	Comments, suggested edits, approve/reject claims, request changes.
Needs evidence	At least one claim lacks support.	Attach evidence, rewrite claim, downgrade confidence, remove claim.
Approved	All required claims/sections approved.	Export, publish, archive, add addendum.
Published	Final shared artifact.	View, export, supersede with new version; edits require new draft version.
Superseded	Older version replaced by a newer approved version.	Read-only except admin restore.

4. Complete Functional Requirements

4.1 Enterprise workspace

- Multi-tenant organizations with strict tenant isolation.
- Workspaces for teams, clients, funds, strategies, investigations, or departments.
- Projects/cases with assigned users, sources, reports, notes, graph views, tasks, and watchlists.
- Role-based and attribute-based access controls over entities, sources, reports, evidence, and exports.
- SSO with SAML/OIDC, SCIM provisioning, MFA, service accounts, API keys, session controls, and audit logs.
- Collaboration: comments, mentions, tasks, annotations, shared saved searches, evidence bookmarks, report review workflow, and approval chains.
- Admin controls: source credentials, connector schedules, model providers, skill registry, export policies, data retention, and billing/usage.

4.2 Entity intelligence

- Global entity search with fuzzy matching, identifier matching, aliases, filters, and source facets.
- Entity profile for companies, people, funds, PACs, nonprofits, agencies, bills, regulations, contracts, court cases, sanctions, and securities.
- Entity identifiers: CIK, ticker, LEI, EIN, UEI, CAGE, OpenCorporates ID, FEC IDs, LDA IDs, FARA IDs, RIN, docket ID, award ID, court ID, ISIN/CUSIP where licensed.
- Graph edges: ownership, investments, officers, board, lobbying, donations, contracts, grants, regulation, public comments, OIRA meetings, lawsuits, sanctions, exclusions, mentions, same-as, and related-party links.
- Timeline events from filings, trades, awards, rules, comments, meetings, court filings, sanctions, earnings, and market events.
- Related-party discovery with ranking by evidence strength, recency, frequency, relationship type, amount, and graph centrality.

4.3 Investor intelligence

- Analyze Stock input supports ticker, company name, CIK, ISIN/CUSIP if licensed, or portfolio position.
- Financial statements: income statement, balance sheet, cash flow, quarterly/annual, TTM, restatement awareness, segment extraction where available.
- Valuation: multiples, peer comps, DCF, reverse DCF, scenario valuation, sensitivity tables, margin of safety, sum-of-the-parts later.
- Technical analysis: moving averages, VWAP, RSI, MACD, Bollinger Bands, ATR, volume, relative volume, support/resistance, trend state, volatility regime.
- Catalysts: earnings, guidance, SEC filings, insider transactions, 13D/G, 13F changes, contracts, grants, Federal Register/rule changes, litigation events, news spikes.
- Risk: valuation risk, liquidity risk, leverage, dilution, customer concentration, procurement concentration, regulatory exposure, litigation, sanctions/exclusions, governance, insider selling, short interest/options if licensed.
- Portfolio: CSV import, position mapping, watchlists, P/L, exposure, concentration, beta, volatility, correlation, catalyst calendar, regulatory exposure, and portfolio memo.

4.4 Public-record intelligence

- SEC filings and facts: company submissions, XBRL, ownership, insider trading, Form D, proxy relationships, 13F, 13D/G.

- Political and lobbying: FEC, LDA, FARA, lobbying issues, clients, lobbyists, registrants, agencies, bills, contributions where allowed.
- Regulation: Congress.gov, GovInfo, Federal Register, Regulations.gov, eCFR, Unified Agenda, RegInfo/OIRA meetings.
- Procurement: USAspending awards, SAM.gov public opportunities/entities/exclusions, subawards, award amounts, NAICS/PSC, agencies.
- Legal and risk: CourtListener/RECAP, OFAC, OpenSanctions, SAM exclusions, enforcement sources later.
- Corporate identity: OpenCorporates, GLEIF LEI, state registries later, Companies House/international later.
- Nonprofit/foundation: IRS Form 990/990-PF officers, grants, related organizations, think tanks, foundations, nonprofit networks.

4.5 Report engine

- Report templates: stock analysis, investor memo, initiating coverage, earnings update, DCF valuation, company intelligence, government exposure, lobbying influence, procurement risk, regulatory exposure, litigation risk, related parties, portfolio review, proposal/pitch.
- Evidence bundle builder that gathers source records, normalized facts, graph paths, financial metrics, charts, and source snippets.
- Claim generator that creates factual and analytical claims only from evidence or explicitly marked assumptions.
- Citation verifier that blocks unsupported or stale claims and flags contradictions.
- Editable Review Workspace with section-level enhancement, comments, tracked changes, model refresh, and approval workflow.
- Exports: PDF, Word, PowerPoint, Excel model, Markdown, HTML, JSON, and evidence appendix CSV.

4.6 Alerting and monitoring

- Watchlists for ticker, company, person, fund, agency, bill, docket, rule, contractor, lobbyist, keyword, portfolio, and case.
- Alert triggers: new SEC filing, Form 4, 13D/G, 13F, Form D, LDA, FARA, bill action, Federal Register item, public comment, OIRA meeting, contract award, SAM opportunity, court case, sanctions hit, nonprofit grant, related party, price movement, volume spike, earnings event, news spike, valuation threshold.
- Delivery: in-app, email, Slack, Teams, webhook, API, and mobile push later.
- Alert throttling, dedupe, severity levels, source confidence, and alert-to-report conversion.

5. Public Records, Market Data, and Financial Source Connectors

Connectors must implement a standard interface: discover, fetch, persist raw, parse, normalize, evidence-link, entity-resolve, publish events, and monitor freshness. The system should prefer official APIs and bulk data over scraping. Official data source examples include SEC EDGAR APIs for submissions and XBRL data [R1], OpenFEC REST API [R2], LDA.gov REST API [R3], and USAspending API endpoints [R4].

5.1 Connector priority matrix

Priorit y	Source	Purpose	Storage strategy	Connector strategy
P0	SEC EDGAR / data.sec.gov	Filings, XBRL facts, ownership, insider trades, Form D, proxy relationships.	Raw JSON/HTML/XML/PDF in object lake; normalized filings/facts in Postgres; analytics in Parquet/warehouse.	Official API + filing downloads; strict rate limits; CIK/ticker cache.
P0	FEC	Campaign finance, candidates, committees, PACs, contributions, disbursements.	Raw API/bulk in lake; normalized committees/contributions in Postgres/lakehouse.	OpenFEC API + bulk; restricted-use tagging for contributor data.
P0	LDA.gov	Lobbying clients, registrants, lobbyists, issues, agencies, bills.	Raw API records in lake; edges in graph; facts in warehouse.	Official REST API; link bills to Congress.gov; issue taxonomy.
P0	Congress.gov	Bills, actions, sponsors, cosponsors, committees.	Raw JSON in lake; normalized bill/member tables.	Official API; incremental by congress/session/date.
P0	Federal Register	Proposed/final rules, notices, executive actions, RINs, CFR refs.	Raw JSON/PDF/text in lake; rule/docket facts in Postgres.	Official API; daily polling by agency/date.
P0	USAspending	Contracts, grants, loans, recipients, agencies, subawards.	Raw API/bulk in lake; award fact tables in warehouse.	API + bulk monthly; use award IDs for idempotency.
P0	OpenCorporates + GLEIF	Company IDs, registry data, LEIs, parent/child, crosswalks.	Raw profiles in lake; identifiers in Postgres.	API enrichment on-demand and scheduled refresh for watched entities.
P0	OpenSanctions/OFAC/SAM exclusions	Sanctions, PEPs, exclusions, risk screening.	Raw lists in lake; screening index in Postgres/OpenSearch.	Bulk list sync; fuzzy screening; source-specific verification.
P1	Regulations.gov	Dockets, comments, attachments, supporting docs.	Attachments in object lake; comments in OpenSearch and lakehouse.	API key; comment text extraction; organization matching.
P1	eCFR	Regulatory text, point-in-time sections, diffs.	Versioned CFR text in lake; diff tables in warehouse.	API plus diff jobs; partition by title/part/date.
P1	RegInfo/OIRA	Unified Agenda, RINs, review status, meetings.	Raw pages/records in lake; meeting/entity edges in graph.	API/scrape where permitted; careful source contracts.
P1	CourtListener/RECAP	Opinions, dockets, parties, litigation risk.	Raw metadata/PDFs in lake; search index and edges.	API with scheduled monitoring for watched entities.
P1	IRS Form 990	Nonprofit/foundation officers, grants, related orgs.	Raw XML in lake; grant/officer fact tables.	Bulk XML parser; annual/monthly refresh.
P1	FARA	Foreign principals, registrants, filings, political activities.	Raw filings in lake; FARA edges in graph.	Bulk/API ingestion; evidence-heavy parser.
P1	Market/financial data	Prices, financial statements, estimates, transcripts, options/short interest.	Provider cache, time-series store, warehouse, and licensed-source controls.	Provider abstraction; use free/SEC data first, paid APIs by tier.

5.2 Financial data providers and alternatives

Provider type	Examples	Use	Cost-control recommendation
Official free data	SEC XBRL/company facts, SEC filings, FRED later	Fundamentals, filings, historical statements.	Default source for U.S. public fundamentals; compute derived metrics internally.
Market data API	Polygon/Massive, Finnhub, Financial Modeling Prep, Nasdaq Data Link	Prices, bars, corporate actions, transcripts, ratios, estimates depending on license.	Use provider abstraction and cache aggressively; tier real-time data only for paid users.
Premium terminals/research	LSEG, S&P Capital IQ, FactSet, PitchBook, Morningstar, Daloopa	Enterprise-grade estimates, transcripts, private-company data, research, models.	Do not ingest unrestrictedly; call via licensed connectors and store only contractually permitted derivatives.
User uploads	CSV, Excel, PDF, data-room files	Portfolio positions, custom peer sets, assumptions, private diligence files.	Store tenant-scoped; classify sensitivity; do not train global models on user data.
Public news/event APIs	GDELT, Media Cloud, RSS, press releases	Narratives, catalysts, media spikes, event monitoring.	Use alerts and entity extraction; retain text only as license allows.

5.3 GitHub repos and tools to reuse

Area	Repos/tools	Recommended use
SEC	dgunning/edgartools; jadchaar/sec-edgar-downloader; sec-edgar-api	Filing download, structured filing parsing, XBRL and SEC data helpers.
FEC	fecgov/openFEC; asg017/libfec; esonderegger/fecfile	Reference API implementation, raw .fec parsing, bulk file processing.
LDA	Lobbying-Disclosure/lobbyr; LobbyViewPythonPackage	API reference and enrichment for lobbying filings and bill links.
FARA	simonw/fara-history/fara-datasette patterns	Bulk data ingestion and SQLite/Datasette workflow inspiration.
Congress	LibraryOfCongress/api.congress.gov; unitedstates/congress; congress-legislators	Bill/member/committee identity and historical data.
Regulatory	usnationalarchives/federalregister-api-core; ecfr-analyzer; regulatory_data_repository	Federal Register ingestion, eCFR diffs, Unified Agenda data.
Spending	fedspendingtransparency/usaspending-api	Reference backend and endpoints for federal spending data.
Courts	freelawproject/courtlistener-api-client	CourtListener/RECAP ingestion and search.
Sanctions	opensanctions/opensanctions	Sanctions/PEP data aggregation and entity resolution patterns.
Finance skills	anthropics/financial-services	Reference agents, plugins, skills, and data connectors for finance workflows.

6. Claude Financial Skills and Managed-Agent Integration

The platform should implement Claude's newly released financial-services agents and skills as an optional but first-class analysis layer. Anthropic's financial-services repository provides reference agents, skills, and data connectors for investment banking, equity research, private equity, and wealth management; the same source can be installed as Claude Cowork plugins or deployed through Claude Managed Agents behind a custom workflow engine [R5]. Anthropic also announced ten ready-to-run financial-services agent templates and states that users remain in the loop reviewing and approving Claude's work before it is used or acted on [R6].

6.1 Supported Claude finance agents

Agent	Platform use in this product	Primary artifacts
Pitch Builder / Pitch Agent	Build proposal/pitch workflows, target lists, comps, client-ready decks, and cover notes.	PowerPoint deck, comps model, proposal memo, cover email draft.
Meeting Preparer	Create entity/counterparty briefings before investor, client, diligence, or board meetings.	Briefing pack, agenda, talking points, open questions.
Market Researcher	Analyze sectors, themes, issuers, competitive landscape, peer comps, and idea shortlists.	Market map, peer list, thesis brief, catalysts.
Earnings Reviewer	Read transcripts and filings, update model assumptions, flag thesis-relevant changes.	Earnings note, model update, guidance change table, risk flags.
Model Builder	Create and maintain DCF, LBO, 3-statement, comps, and scenario models in Excel or internal model format.	Excel workbook, assumptions JSON, sensitivity tables.
Valuation Reviewer	Check valuations against comps, methodology, firm standards, and report assumptions.	Valuation review memo, methodology exceptions, approval checklist.
KYC Screener	Support public-record KYB/KYC-style packs and diligence escalations where licensed and appropriate.	Entity file, risk checklist, missing docs, escalation notes.

6.2 Supported Claude finance skills

Anthropic's 2025 finance update described six finance skills: comparable-company analysis, discounted cash flow models, due-diligence data packs, company teasers/profiles, earnings analyses, and initiating coverage reports [R7]. The platform should expose those as auditable workflow steps.

Skill	Inputs	Outputs	Platform controls
Comparable company analysis	Target company, peer set, fiscal periods, metrics, market data, user overrides.	Peer table, multiples, percentile ranges, valuation summary.	Source map for every metric; user can edit peer set; refreshable.
Discounted cash flow	Historical financials, projections, WACC, terminal assumptions, tax/capex/NWC assumptions.	DCF model, enterprise value, equity value, sensitivity tables, scenarios.	Assumptions versioned; reviewer approval required for final reports.
Due diligence data pack	Data-room docs, filings, contracts, customer lists, financial files, public records.	Structured diligence spreadsheet, issue log, red flags, document index.	Tenant-scoped files; confidentiality labels; evidence refs.
Company teaser/profile	Company profile, financials, market data, products, customers, comps, public records.	Short profile, pitchbook pages, buyer-list summary.	No unsupported claims; editable proposal mode.
Earnings analysis	Transcript, 10-Q/8-K, financial model, estimates if licensed, prior guidance.	Earnings note, metrics table, guidance changes, management commentary.	Transcript/source citations; compare to prior periods.
Initiating coverage report	Industry overview, company deep-dive, financials, valuation, catalysts, risks.	Full coverage report with model, charts, thesis, risks, appendix.	Human review and citation verification required.

6.3 Finance Skills Gateway architecture

Implement a provider-agnostic FinanceSkillsGateway. Do not hard-code the product to one LLM provider. The gateway should route workflows to Anthropic Claude financial services when enabled, or to internal skills/templates when not enabled. All invocations should be logged, all input artifacts should be hashed, and all outputs should be stored as versioned artifacts.

```
services/finance_skills_gateway/  
registry.py # Skill and agent catalog, allowlist, version pins  
providers/  
anthropic_managed_agents.py  
claude_cowork_plugin_adapter.py  
internal_finance_skills.py
```

```

workflows/
comps.py
dcf.py
earnings.py
initiating_coverage.py
due_diligence_pack.py
pitch_builder.py
valuation_review.py
artifacts/
excel_writer.py
powerpoint_writer.py
report_bundle.py
security/
sandbox_policy.py
prompt_redaction.py
data_access_policy.py
tests/
fixtures/

```

6.4 Skill invocation object

```

SkillRun {
  id: uuid
  tenant_id: uuid
  workspace_id: uuid
  provider: 'anthropic' | 'internal' | 'other'
  skill_name: 'dcf' | 'comps' | 'earnings' | 'initiating_coverage' | ...
  provider_skill_ref: string
  version: string
  input_artifacts: [ArtifactRef]
  allowed_sources: [SourceId]
  prompt_hash: sha256
  output_artifacts: [ArtifactRef]
  evidence_refs: [EvidenceRef]
  cost_estimate: decimal
  actual_cost: decimal
  status: queued | running | completed | failed | needs_review
  created_by: user_id
  reviewed_by: user_id | null
  created_at: timestamp
  completed_at: timestamp | null
}

```

6.5 Integration paths

Path	How it works	When to use	Risk/cost controls
Claude Managed Agents	Deploy Anthropic's managed-agent cookbooks behind the platform's workflow engine, passing controlled data bundles and receiving artifacts.	Best for production-grade asynchronous workflows, nightly runs, and enterprise audit trails.	Credential vault, per-tool permissions, audit logs, data-scoped bundles, cost budgets.
Claude Cowork / Claude Code plugins	Use the financial-services marketplace plugins for analyst-in-the-loop desktop workflows and internal developer/operator workflows.	Best for internal analysts, pilots, and template customization.	Human-in-loop, no direct trading, manual export checks, plugin allowlist.
Internal skill fallback	Implement core comps/DCF/earnings/report templates directly using platform data and a model-agnostic LLM gateway.	Best when Claude financial-services tools are unavailable, unlicensed, or need strict private deployment.	Same schema as Claude outputs; same evidence verification.

6.6 Product workflows using Claude skills

Workflow	Claude skills/agents	Required platform modules
Analyze Stock - long-term investor	Market Researcher + Model Builder + DCF + Comps + Earnings Reviewer + Valuation Reviewer.	SEC facts, market data, peer set, valuation engine, report editor, citation verifier.
Day trader catalyst brief	Market Researcher + Earnings Reviewer; internal technical analysis module.	Market data, alerts, news/filings, technical indicators, concise brief template.
Investment committee memo	Model Builder + Valuation Reviewer + Initiating Coverage + Due Diligence Pack.	Financial data, entity graph, evidence bundle, approval workflow.
Proposal / pitch deck	Pitch Builder + Company Teaser/Profile + Comps.	Report-to-deck exporter, proposal editor, client template library.
Earnings update	Earnings Reviewer + Model Builder.	Transcript source, earnings release, 10-Q/8-K, prior model, guidance table.

Workflow	Claude skills/agents	Required platform modules
Related-party diligence	Market Researcher + KYC Screener + Due Diligence Pack.	Entity graph, public records, sanctions, court records, evidence review.

6.7 Security and governance for Claude finance integration

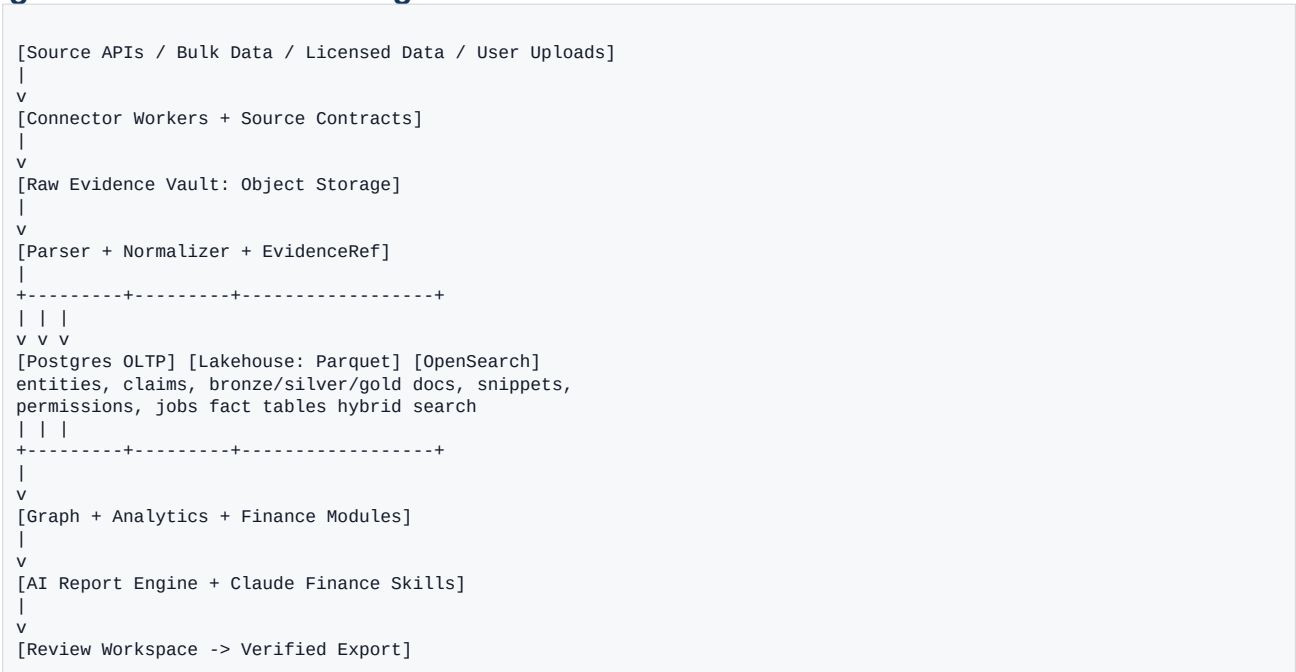
- Use a SkillRegistry allowlist; no arbitrary skill packages in production.
- Pin skill/plugin versions or Git commit hashes; scan skill instructions and scripts before enabling.
- Run any executable scripts in a sandbox with no unrestricted network or file access.
- Pass scoped evidence bundles, not entire tenant databases, to skill runs.
- Log prompts, tool calls, data sources used, model versions, generated artifacts, costs, and review status.
- Require human review before any externally shared investment report, proposal, valuation, or due-diligence memo.
- Never auto-execute trades, submit filings, approve onboarding, post to ledgers, or bind risk from Claude outputs.
- Enforce the same citation verifier on Claude-generated text as on internally generated report text.

7. Data Architecture: OLTP, Lake, Warehouse, Search, Graph, Vector

7.1 Logical storage layers

Layer	Primary data	Recommended technology	Reason
Operational OLTP	Users, orgs, permissions, entities, identifiers, claims, reports, jobs, connector state.	Postgres / Aurora / Cloud SQL / Supabase enterprise / Neon enterprise.	Relational integrity, transactions, application queries, row-level permissions.
Raw evidence lake	Raw API JSON, XML, CSV, PDFs, HTML, filings, attachments, transcripts, screenshots where allowed.	S3/GCS/Azure Blob/Cloudflare R2/MinIO with immutable paths.	Lowest-cost durable storage; supports replay, audits, and legal defensibility.
Lakehouse tables	Parsed records, normalized event/fact tables, append-only historical data.	Parquet + Iceberg preferred; Delta Lake if Databricks-heavy.	Open formats, column pruning, partitioning, schema evolution, warehouse independence.
Warehouse/OLAP	Aggregated facts, dashboards, portfolio analytics, contract/lobbying/regulatory trends.	ClickHouse, BigQuery, Snowflake, Athena/Trino, Databricks SQL.	Fast analytics, ad hoc queries, BI, cost/performance separation.
Search	Documents, filings, comments, reports, snippets, source text.	OpenSearch/Elasticsearch; Typesense/Meilisearch for smaller deployments.	Full-text, faceting, highlighting, hybrid retrieval.
Vector	Embeddings for high-value document chunks, evidence bundles, entity summaries.	OpenSearch vector, pgvector, Qdrant, Weaviate, Pinecone.	Semantic retrieval and RAG; choose based on scale/cost.
Graph	Entity relationships and pathfinding projections.	Start in Postgres edge tables; add Neo4j/ArangoDB/Neptune later.	Avoid premature graph infra cost; move when traversals require it.

7.2 High-level architecture diagram



7.3 Bronze, Silver, Gold data layers

Layer	Contents	Rules	Storage
Bronze	Raw or lightly decoded source records, retaining original field names and payloads.	Append-only; never overwrite; dedupe by hash and source_record_id.	Object store plus Parquet for large structured feeds.
Silver	Parsed and normalized source-specific tables with consistent datatypes and basic cleaning.	Idempotent transforms; schema evolution; partition by source/date; retain source IDs.	Iceberg/Delta/Hudi tables in lakehouse.
Gold	Business-ready facts: entity events, award facts, lobbying facts, financial metrics, risk flags, rollups.	Optimized for dashboard/report queries; materialized views; source lineage required.	Warehouse/ClickHouse/Postgres materialized tables depending on query pattern.

7.4 Recommended initial physical deployment

- Postgres: operational system of record, partitioned audit/relationship/event tables, read replica for analytics-lite and API reads.
- Object storage: raw evidence bucket with tenant/source/date/entity prefixes; lifecycle policies; hash-based dedupe.
- Lakehouse: Parquet + Iceberg catalog for high-volume datasets; start with simple partitioning and add compaction jobs early.
- ClickHouse: event/fact analytics, alert queries, time-series, large aggregations, and fast dashboards.
- OpenSearch: full-text search and hybrid semantic search across filings, comments, reports, and evidence snippets.
- pgvector initially for small embeddings; OpenSearch vector or Qdrant for large-scale semantic search.
- Graph layer in Postgres for V1; add Neo4j only when pathfinding latency or graph algorithms exceed relational capacity.

7.5 Why open table formats matter

Apache Iceberg describes itself as a high-performance format for huge analytic tables and enables engines such as Spark, Trino, Flink, Presto, Hive, and Impala to work with the same tables [R8]. Delta Lake is an open-source storage framework for lakehouse architectures with support across multiple compute engines [R9]. Apache Parquet is a column-oriented open storage format designed for efficient storage and retrieval [R10]. The platform should use these open formats to avoid locking long-term historical data into one warehouse vendor.

8. Scaling and Cost-Minimization Strategies

8.1 Cost risks to avoid

- Loading every raw record into an expensive warehouse before it is needed.
- Embedding every document chunk regardless of usage or value.
- Polling APIs too frequently without change detection, backoff, or watchlist-driven priority.
- Using paid market-data APIs for free official data that can be computed from SEC facts.
- Creating huge unpartitioned warehouse tables that scan entire datasets on every query.
- Duplicating raw documents across object storage, warehouse binary columns, and search indexes.
- Running large LLM report jobs repeatedly when only a small source update changed.
- Self-hosting complex infrastructure too early if the team lacks ops capacity, or staying on expensive managed tools after workloads stabilize.

8.2 Hot/warm/cold data strategy

Tier	Data	Target storage	Retention / cost strategy
Hot	Active entities, current reports, user workspaces, latest filings, latest market data, watchlist events.	Postgres, Redis, OpenSearch, ClickHouse hot tables.	Fast access; indexes; short retention for caches; active alerts.
Warm	Historical normalized events, quarterly filings, lobbying records, contracts, dockets, report evidence bundles.	Lakehouse Parquet/Iceberg, ClickHouse/Snowflake/BigQuery as needed.	Partitioned, compressed, compacted; queryable for analytics.
Cold	Old raw files, large PDFs, superseded source records, low-value attachments.	Object storage lifecycle tiers: S3 IA/Glacier or equivalents.	Keep hashes and metadata hot; retrieve content only when needed.

8.3 API cost strategy

Pattern	Implementation	Expected benefit
Official API first	Prefer SEC, FEC, LDA, USAspending, Congress.gov, Federal Register, Regulations.gov, eCFR, CourtListener, IRS bulk, OpenSanctions.	Reduces reliance on paid vendors and improves defensibility.
On-demand enrichment	Only call expensive paid APIs when a user opens an entity, runs a report, or the entity is on a watchlist.	Avoids paying to enrich the entire universe.
Connector priority queue	Prioritize active tenants, portfolios, watchlists, and scheduled reports over cold universe sync.	Compute and API calls follow revenue/value.
TTL cache	Cache provider responses by endpoint/source/license constraints; expire based on data volatility.	Reduces repeated paid API calls.
Bulk before endpoint loops	Use bulk downloads for large official sources, then incremental endpoints for updates.	Reduces request volume and improves reliability.
Change detection	Use last_modified, accession numbers, filing dates, checksums, ETags, or source-specific cursors.	Avoids refetching unchanged records.
User-visible cost budgets	Estimate report cost before deep runs; set workspace/model/API budgets.	Prevents runaway LLM and data-provider spend.

8.4 Warehouse query cost strategy

BigQuery pricing separates compute and storage, with compute driven by queries and bytes processed [R11]. Amazon Athena charges by data scanned and can reduce cost by compression, partitioning, and columnar formats such as Parquet [R13]. Snowflake uses a consumption-based model based on compute credits, storage, and data transfer [R12]. Therefore the platform should enforce query-cost controls regardless of warehouse provider.

- Use columnar Parquet with ZSTD or Snappy compression for lakehouse tables.
- Partition high-volume tables by source, date, tenant, and sometimes entity type. Avoid over-partitioning by high-cardinality entity IDs unless query patterns justify it.

- Cluster/sort by entity_id, source_record_date, ticker, agency_id, award_id, or docket_id depending on workload.
- Create gold rollups for report dashboards: monthly lobbying by client, awards by recipient/agency, filings by company/date, price metrics by ticker/day.
- Set maximum bytes billed or query guardrails where supported.
- Use materialized views for common expensive joins.
- Separate analyst ad hoc sandboxes from production report queries.
- Schedule heavy transforms off-peak and precompute report bundles for watched entities.
- Monitor slow queries and cost by tenant, user, report, connector, and source.

8.5 Object lake cost strategy

- Never store duplicate raw documents with different file names; hash content and reference it from multiple source records if identical.
- Compress JSON/CSV source dumps with gzip/zstd; convert large structured data to Parquet for analytics.
- Use object prefixes: tenant/source/year/month/day/entity/source_record_id.
- Separate public/global raw data from tenant-private uploads.
- Apply lifecycle policies: hot 30-90 days, warm 12-24 months, cold/archive for long retention.
- Keep lightweight metadata hot even when raw binary content moves to archive.
- For Cloudflare R2 or similar low-eegress storage, evaluate compatibility and governance before choosing over AWS/GCP/Azure native storage.

8.6 Search and embedding cost strategy

- Index metadata and high-value text first; do not index every raw attachment immediately.
- Chunk intelligently by document structure: SEC item sections, public comment body, court docket entry, report section, transcript Q&A;., not arbitrary fixed token chunks only.
- Embed only documents in watched entities, active reports, high-value sources, or user-requested research packs.
- Deduplicate chunks by hash before embedding.
- Cache embeddings by model version and chunk hash.
- Use lexical search first, then semantic reranking; reserve large-context LLM calls for final report synthesis.
- Create entity summaries and evidence bundles to reduce repeated full-document retrieval.

8.7 LLM and report-generation cost strategy

- Use tiered models: small/cheap models for classification, extraction, routing, and dedupe; premium models for final analysis and complex financial reports.
- Use retrieval budgets: fixed maximum evidence documents, chunks, and tokens per section.
- Generate reports section-by-section and cache section outputs by evidence bundle hash.
- When new data arrives, run incremental updates instead of regenerating the entire report.
- Store report assumptions, claims, and generated artifacts separately so edits do not force all sections to rerun.
- Make deep Claude financial-skill runs explicit user actions with estimated cost and expected artifacts.
- Batch daily/weekly portfolio briefs rather than running each position independently when possible.

8.8 Scaling thresholds and migration triggers

Component	Start	Trigger to scale	Next step
Postgres	Single primary + read replica.	CPU > 60% sustained, slow relationship queries, table growth > 500M rows, write contention.	Partition event/edge tables, add replicas, move analytics to warehouse/ClickHouse, consider Citus/Aurora.
Graph	Postgres edge tables.	Path queries > 5s, recursive traversals heavy, graph algorithms needed.	Materialized graph projections, then Neo4j/ArangoDB/Neptune for selected workloads.
Search	Managed OpenSearch small cluster.	Index > several TB, query latency > 1s, embedding workloads heavy.	Shard by tenant/source/date, hot/warm nodes, vector index separation.
Warehouse	ClickHouse or serverless warehouse.	Dashboards/report aggregations slow or query cost unpredictable.	Introduce gold tables, materialized views, capacity reservations or dedicated clusters.
Lake	Object store + Parquet.	Schema evolution, concurrent writers, table snapshots needed.	Adopt Iceberg/Delta catalog, compaction, snapshot management.
Workers	Single queue group.	Backlogs exceed SLA or source refresh windows.	Separate queues by source priority and workload type; autoscale workers.
LLM	Synchronous report calls for early demos.	Reports exceed timeout/cost or need review artifacts.	Async report jobs, evidence bundles, cost budgets, cached sections.

9. Data Lake, Warehouse, and API Alternatives

9.1 Recommended option: Hybrid cost-aware lakehouse

The recommended option for long-term cost control is a hybrid lakehouse. Keep raw evidence in object storage, normalize facts into Postgres and Parquet/Iceberg, use ClickHouse or a serverless query engine for analytics, and add a managed warehouse only for workloads that require it. This avoids paying warehouse prices for cold raw evidence while still enabling enterprise analytics.

Component	Choice	Rationale
Object storage	S3 / GCS / Azure Blob / R2 / MinIO	Low-cost durable raw evidence storage and lifecycle policies.
File format	Parquet	Columnar storage, compression, and efficient analytics [R10].
Table format	Iceberg preferred; Delta if Databricks chosen	Schema evolution, snapshots, engine interoperability, large analytic tables [R8][R9].
Query	DuckDB for local/dev; Athena/Trino for lake; ClickHouse for fast analytics	Minimize fixed costs; scale based on query patterns.
Warehouse	BigQuery/Snowflake optional	Use when enterprise governance, BI integrations, or customer requirements justify managed cost.

9.2 Option A - Lowest-cost open-source-first

Area	Stack	Pros	Cons
OLTP	Postgres + pgvector	Low cost, flexible, familiar.	Needs tuning and ops as scale grows.
Lake	MinIO/S3/R2 + Parquet + Iceberg	Cheap storage, open formats, replayable.	Requires data engineering discipline.
Analytics	DuckDB + Trino/Athena + ClickHouse self-managed	Excellent cost/performance for disciplined teams.	More ops burden; careful security/governance needed.
Search	OpenSearch self-managed	Powerful, flexible, no per-document vendor tax.	Cluster ops can be complex.
Graph	Postgres edges first	Avoids early graph cost.	Complex graph algorithms later need a dedicated graph DB.

Best for: early company with strong engineering discipline, cost sensitivity, and willingness to operate infrastructure.

9.3 Option B - Serverless cloud-first

Area	AWS version	GCP version	Pros/cons
OLTP	RDS/Aurora Postgres	Cloud SQL/AlloyDB	Managed reliability; higher cloud cost.
Lake	S3 + Glue Catalog + Iceberg	GCS + BigLake/Iceberg	Strong native integration; cloud lock-in risk.
Query	Athena + Glue	BigQuery external/native tables	Low ops; costs can spike with scans.
Search	Amazon OpenSearch	Elastic/OpenSearch managed	Managed search; vector costs must be monitored.
Workers	ECS/EKS/Lambda	Cloud Run/GKE	Autoscaling; cold-start and networking complexity.

Best for: small-to-mid teams wanting minimal operations and faster enterprise reliability.

9.4 Option C - Managed warehouse enterprise

Warehouse	Strengths	Risks	Use when
Snowflake	Enterprise governance, sharing, mature BI ecosystem, compute/storage separation.	Consumption costs can surprise; raw evidence storage in warehouse is expensive.	Enterprise customers demand Snowflake, cross-cloud, data sharing, governed SQL.
BigQuery	Serverless, strong GCP integration, large-scale SQL, native ML/BI features.	Bytes-processed costs require guardrails; GCP-centric.	Team is on GCP or wants low-ops serverless analytics.
Databricks	Lakehouse, Spark, ML, Delta, governance with Unity Catalog, notebook ecosystem.	Can be expensive and ops-heavy if not governed.	Need large-scale Spark, ML, complex ETL, and Delta ecosystem.
ClickHouse Cloud	Fast analytics for events/time-series/logs; excellent dashboard latency.	Not a general-purpose governance warehouse; SQL patterns differ.	High-volume events, alerts, time-series, and fast analytical dashboards.

9.5 Option D - API-first / cache-light alternative

An API-first architecture avoids ingesting large datasets until a user asks for them. This can be cheaper early, but it creates latency, dependency, and reproducibility risks. It is useful for paid data vendors, low-frequency sources, and sources with strict redistribution rules.

Use API-first when	Use lake ingestion when
Data is expensive, licensed, or contractually restricted.	Data is public, bulk-accessible, and useful for many users.
User requests are rare or unpredictable.	Queries recur across users, reports, alerts, and dashboards.
The provider offers strong search/filter endpoints.	The source lacks good filtering or has fragile APIs.
Latency is acceptable.	Reports need reproducible evidence and fast repeated analysis.
Storage rights are limited.	The data can be retained in raw form with source terms.

9.6 Recommended phased data strategy

- 1 Phase 1: Postgres + object storage + OpenSearch + simple Parquet export. Use APIs on demand and cache raw records.
- 2 Phase 2: Add lakehouse tables for high-volume sources: SEC filings metadata, FEC, LDA, USAspending, Regulations.gov comments, market bars.
- 3 Phase 3: Add ClickHouse for alerts, dashboards, time-series, and high-volume aggregations.
- 4 Phase 4: Add Iceberg catalog and compaction jobs when large tables need schema evolution and consistent snapshots.
- 5 Phase 5: Offer managed warehouse integration for enterprise customers: Snowflake/BigQuery/Databricks exports or native warehouse deployment.
- 6 Phase 6: Move selected graph workloads to Neo4j/ArangoDB/Neptune when pathfinding and graph algorithms justify the cost.

10. Data Model and API Contracts

10.1 Core relational model

```
organizations
workspaces
users
roles
permissions
projects
cases
watchlists
portfolios
entities
entity_aliases
entity_identifiers
relationships
relationship_evidence
source_registry
source_credentials
source_runs
source_records
raw_documents
evidence_refs
claims
claim_evidence
reports
report_sections
report_versions
report_comments
review_tasks
skill_runs
artifacts
alerts
alert_events
audit_logs
entity_merge_events
data_quality_issues
```

10.2 Entity object

```
Entity {
  id: uuid
  tenant_id: uuid | null # null for global public entity, tenant id for private entities
  type: enum(Person, Company, Fund, Security, Agency, Bill, Regulation, Docket,
  ContractAward, GrantAward, CourtCase, Nonprofit, PAC, Committee,
  LobbyingFirm, Lobbyist, SanctionsRecord, Document, Report, Other)
  canonical_name: string
  aliases: [string]
  identifiers: [Identifier]
  jurisdictions: [string]
  description: text
  confidence: decimal
  created_at: timestamp
  updated_at: timestamp
}
```

10.3 Relationship object

```
Relationship {
  id: uuid
  tenant_id: uuid | null
  subject_entity_id: uuid
  predicate: enum(OWNS, INVESTED_IN, OFFICER_OF, BOARD_MEMBER_OF, LOBBIED_FOR,
  LOBBIED_ON, DONATED_TO, SUBMITTED_COMMENT, ATTENDED_OIRA_MEETING,
  SPONSORED_BILL, RECEIVED_CONTRACT, RECEIVED_GRANT, PARTY_TO_CASE,
  SANCTIONED_BY, EXCLUDED_BY, SAME_AS, MENTIONED_IN, RELATED_TO)
  object_entity_id: uuid
  start_date: date | null
  end_date: date | null
  amount: decimal | null
  currency: string | null
  confidence: decimal
  evidence_refs: [uuid]
  source_record_ids: [uuid]
  extraction_method: enum(official_api, bulk_file, parser, llm_extraction, analyst, imported)
  review_status: enum(auto, needs_review, approved, rejected)
```

```
}
```

10.4 EvidenceRef object

```
EvidenceRef {
  id: uuid
  source_name: string
  source_record_id: uuid
  raw_document_id: uuid | null
  source_url: string | null
  source_record_identifier: string | null # accession no, docket id, award id, filing id, etc.
  document_type: string
  published_at: timestamp | null
  retrieved_at: timestamp
  excerpt: text | null
  field_path: string | null
  page_number: integer | null
  char_start: integer | null
  char_end: integer | null
  hash: sha256
}
```

10.5 Claim object

```
Claim {
  id: uuid
  report_id: uuid | null
  entity_id: uuid | null
  claim_text: text
  claim_type: enum(financial, valuation, legal, lobbying, procurement, regulatory,
ownership, political_money, market_signal, related_party, risk, other)
  confidence_label: enum(confirmed, likely, possible, unverified, contradicted)
  evidence_refs: [uuid]
  assumptions: [Assumption]
  contradictions: [ClaimContradiction]
  generated_by: enum(system, analyst, claude_skill, internal_llm, imported)
  review_status: enum(draft, needs_evidence, needs_review, approved, rejected)
  created_by: user_id
  reviewed_by: user_id | null
}
```

10.6 Source contract YAML

```
source: sec_edgar
owner: data-platform
category: public_records
official: true
base_url: https://data.sec.gov
auth: none
rate_limit:
  requests_per_second: 8
  burst: 10
raw_storage: required
raw_storage_path: s3://evidence/public/sec_edgar/{yyyy}/{mm}/{dd}/{record_id}
normalization: required
evidence_required: true
refresh:
  watched_entities: hourly
  universe_metadata: daily
  historical_backfill: batch_manual
restricted_use:
  enabled: false
legal_notes:
  - comply_with_source_terms
  - preserve_user_agent_contact
outputs:
  entities: [Company, Filing, Security, Person]
  relationships: [OFFICER_OF, BOARD_MEMBER_OF, OWNS, INVESTED_IN, MENTIONED_IN]
quality_checks:
  - raw_document_hash_present
  - source_record_id_unique
  - evidence_ref_required
  - parser_version_logged
```

10.7 Public API endpoints

Endpoint	Method	Purpose
/v1/search	GET	Global search across entities, documents, reports, claims, and evidence.
/v1/entities/{id}	GET	Fetch entity profile with identifiers, aliases, summary, and risk flags.
/v1/entities/{id}/relationships	GET	Fetch relationship edges with filters and evidence refs.
/v1/entities/{id}/graph	GET/POST	Graph expansion and pathfinding query.
/v1/entities/{id}/timeline	GET	Chronological events from filings, contracts, lobbying, court, market data.
/v1/analyze/stock	POST	Run investor stock analysis workflow.
/v1/reports	POST	Create report draft from template and evidence bundle.
/v1/reports/{id}/enhance	POST	Enhance or rewrite selected report section.
/v1/reports/{id}/verify	POST	Run citation and claim verification.
/v1/reports/{id}/export	POST	Export PDF/Word/PPT/Excel/HTML/JSON.
/v1/skills/run	POST	Run approved finance skill through FinanceSkillsGateway.
/v1/connectors/{source}/run	POST	Trigger connector run with dry-run, watched, or backfill mode.
/v1/alerts	POST/GET	Create and list alert rules.
/v1/portfolios	POST/GET	Upload/list portfolios and positions.

11. Report Engine, Evidence, Claims, and Review Workflow

11.1 Report generation pipeline

- 1 Resolve entity and scope: ticker/company/fund/person/agency/bill/portfolio; define report type, time horizon, sources, and depth.
- 2 Build Evidence Bundle: gather source records, raw documents, normalized facts, graph paths, financial metrics, market data, and analyst notes.
- 3 Run specialized modules: public-record analysts, financial analysis, graph pathfinding, risk scoring, and Claude finance skills where enabled.
- 4 Generate structured report draft with sections, claim cards, charts, tables, assumptions, model outputs, and source appendix.
- 5 Run citation verifier and contradiction scanner.
- 6 Move report to Review Workspace for editing, enhancement, comments, tracked changes, and approvals.
- 7 After edits, re-run claim verification and model consistency checks.
- 8 Approve final version and export artifacts.

11.2 Evidence bundle schema

```
EvidenceBundle {  
  id: uuid  
  target_entity_id: uuid  
  report_type: string  
  data_cutoff_at: timestamp  
  source_records: [SourceRecordRef]  
  documents: [DocumentRef]  
  financial_metrics: [MetricRef]  
  graph_paths: [GraphPath]  
  market_events: [MarketEvent]  
  assumptions: [Assumption]  
  excluded_sources: [SourceExclusion]  
  quality_warnings: [DataQualityWarning]  
  bundle_hash: sha256  
}
```

11.3 Report templates

Template	Required sections
Stock analysis	Executive summary, business overview, price action, fundamentals, valuation, technicals, earnings, catalysts, risks, public-record intelligence, bull/base/bear, monitoring plan, evidence appendix.
Investor memo	Thesis, market opportunity, financial quality, valuation, management, moat, government/regulatory exposure, risks, catalysts, decision checklist, appendix.
Initiating coverage	Industry overview, company deep dive, financials, valuation framework, peer comps, management, catalysts, risks, estimates, evidence/source appendix.
Government exposure	Agencies, awards, contracts, grants, lobbying, public comments, OIRA meetings, regulations, procurement concentration, policy risks.
Related-party report	Entity map, relationship table, key parties, evidence, relationship strength, timeline, risk/opportunity interpretation, open questions.
Proposal/pitch	Client context, problem, research approach, findings, recommended next steps, deliverables, timeline, team, assumptions, evidence/appendix.

11.4 Section-level enhancement prompts

Action	Prompt behavior	Guardrail
Improve clarity	Rewrite selected section in plain executive language without changing facts.	Must preserve citations and claim IDs.
Add evidence	Search allowed sources for additional support and insert cited facts.	New facts require EvidenceRefs.
Make investor-grade	Add valuation context, catalysts, risks, assumptions, and monitoring metrics.	No unsupported buy/sell recommendation.
Add counterargument	Generate credible bear/alternative case from existing evidence and risk sources.	Label as analysis, not fact unless supported.

Action	Prompt behavior	Guardrail
Create proposal	Convert research into client-facing proposal or pitch structure.	Separate proven findings from proposed work.
Refresh latest	Run update connector jobs for target sources and identify changed sections.	Store new evidence bundle and version delta.
Run Claude finance skill	Send scoped evidence/model bundle to approved Claude skill.	Log SkillRun and verify output claims.

11.5 Claim verification rules

- Every factual claim must have at least one EvidenceRef or must be explicitly labeled as an assumption/opinion.
- Every financial model output must have an assumptions table and data source references.
- Every valuation conclusion must show scenario range, not a single unsupported number.
- Every claim involving wrongdoing, conflicts, sanctions, legal risk, or political influence requires high-confidence evidence and reviewer approval.
- If sources disagree, the report must show the contradiction and avoid a definitive claim unless one source is authoritative.
- If a user manually edits a paragraph and removes citations, the section is marked Needs Evidence.
- AI-generated proposal language cannot upgrade possible/unverified claims into confirmed claims.

12. Security, Compliance, and Governance

12.1 Enterprise security

- SSO/SAML/OIDC, SCIM, MFA, tenant isolation, RBAC/ABAC, field-level permissions, and service accounts.
- Encryption at rest and in transit; secrets manager; key rotation; optional customer-managed keys for enterprise tier.
- Audit logs for searches, report generation, evidence views, exports, skill runs, connector runs, admin changes, entity merges, and approvals.
- IP allowlists, session controls, device policies, export controls, and admin approvals for sensitive exports.
- Private deployment option for regulated clients; VPC peering or private link for enterprise data sources.
- Dependency scanning, container scanning, SBOM, vulnerability management, and penetration testing before enterprise launch.

12.2 Source compliance

- Respect API terms, rate limits, robots restrictions, authentication rules, and data redistribution limits.
- Do not bypass CAPTCHAs, paywalls, logins, or technical access controls.
- Use official APIs/bulk downloads first; scrape only when lawful, permitted, and no official API exists.
- Track source licenses, usage limits, refresh cadence, and restricted fields.
- FEC individual contributor data must be restricted and not used for solicitation or unlawful commercial misuse.
- SAM.gov sensitive data must be separated from public data and never shown in public-facing views unless permitted.
- Licensed market/research providers must be accessed through provider-specific data policies and storage restrictions.

12.3 Financial research controls

- The platform provides research and decision support, not guaranteed financial advice.
- No automatic trade execution in this platform scope.
- Every investment memo must include risk factors, assumptions, confidence labels, and data cutoff date.
- Valuation outputs must show assumptions, sensitivity, and source data.
- Human review is required for externally distributed investment memos, reports, and proposals.
- Claude or other AI financial outputs must remain draft analyst work product until approved.

12.4 Privacy and sensitive-data controls

- Avoid unnecessary exposure of private residential addresses unless clearly relevant and lawfully public in context.
- Separate public officials, business entities, and private persons in data model and UI labeling.
- Support redaction and masking for sensitive fields in exports.
- Do not build private-person surveillance features.
- Tenant-uploaded documents must remain tenant-scoped and excluded from global indexes unless explicitly authorized.

13. Developer Roadmap, Workstreams, and Codex Plan

Codex should be used to accelerate implementation through scoped GitHub issues and pull requests. Codex web can connect to GitHub repositories and create pull requests from its work [R21], and Codex code review can review pull-request diffs and post GitHub reviews [R22]. Codex CLI is a local coding agent that runs in the terminal [R23].

13.1 Repository structure

```
enterprise-intelligence-platform/  
  AGENTS.md  
  README.md  
  docker-compose.yml  
  pyproject.toml  
  package.json  
  apps/  
  api/ # FastAPI backend  
  web/ # Next.js frontend  
  admin/ # Enterprise admin UI  
  worker/ # ingestion, reports, alerts  
  packages/  
  connectors/ # SEC, FEC, LDA, FARA, Congress, etc.  
  ontology/ # entity, relationship, claim schemas  
  entity_resolution/ # deterministic/probabilistic matching  
  graph/ # pathfinding and graph APIs  
  search/ # OpenSearch and vector index services  
  evidence/ # raw docs, evidence refs, source records  
  reporting/ # reports, claims, templates, exports  
  finance/ # fundamentals, valuation, technicals  
  skills_gateway/ # Claude finance skills and internal skills  
  alerts/ # alert rules and delivery  
  portfolios/ # portfolio ingestion and analytics  
  compliance/ # source policies, review, redaction  
  infra/  
  docker/  
  kubernetes/  
  terraform/  
  migrations/  
  docs/  
  architecture/  
  source_contracts/  
  methodology/  
  report_templates/  
  security/  
  tests/  
  unit/  
  integration/  
  fixtures/  
  contract/
```

13.2 AGENTS.md requirements for Codex

```
Rules for all Codex-generated code:  
- Use official APIs and bulk data before scraping.  
- Never bypass CAPTCHAs, authentication, paywalls, robots restrictions, or rate limits.  
- Every connector must store raw records before normalization.  
- Every normalized entity, relationship, metric, and claim must include EvidenceRef.  
- Every connector must include unit tests, fixtures, source_contract.yml, and docs.  
- Every financial output must expose assumptions, source data, scenarios, and risk factors.  
- Do not generate unsupported allegations.  
- Do not merge entities without deterministic identifiers or approved confidence.  
- Treat FEC individual contributor data as restricted-use.  
- Separate public and sensitive SAM.gov data.  
- Claude finance skill outputs are drafts until reviewed and verified.  
- Report edits must trigger citation verification.
```

13.3 Workstreams

Workstream	Scope	Initial deliverables
Enterprise core	Organizations, workspaces, auth, RBAC, audit, projects, cases, admin.	Multi-tenant skeleton, user roles, audit logs, workspace UI.
Evidence and source platform	Source registry, source contracts, raw storage, connector runs, evidence refs.	SourceRecord, RawDocument, EvidenceRef, connector SDK.
Connectors	SEC, FEC, LDA, Congress, Federal Register, USAspending, OpenCorporates, GLEIF, sanctions, etc.	P0 connectors with tests and raw persistence.

Workstream	Scope	Initial deliverables
Entity graph	Entities, identifiers, relationships, resolution, graph API, related-party discovery.	Entity profile, relationship edges, pathfinding, merge queue.
Financial analysis	Market data, fundamentals, valuation, DCF, comps, technicals, portfolio.	Analyze Stock workflow, valuation engine, portfolio import.
Claude skills gateway	Approved Anthropic finance agents/skills, internal fallbacks, artifacts, audit logs.	SkillRegistry, SkillRun, DCF/comps/earnings adapters.
Report engine	Evidence bundles, claims, templates, editor, review, verifier, exports.	Editable report workspace with verification and PDF/Word/PPT/Excel export.
Data platform	Lakehouse, warehouse, search, vector, graph storage, monitoring, cost controls.	Parquet lake, OpenSearch, ClickHouse/warehouse optional, budgets.

13.4 Phase plan

Phase	Goal	Key outputs
0	Architecture and repo foundation	Monorepo, Docker, CI, AGENTS.md, source contract template, base schemas.
1	Enterprise core	Organizations, workspaces, RBAC, audit logs, projects, admin UI.
2	Evidence and source SDK	Raw evidence vault, connector runner, source registry, EvidenceRef.
3	P0 connectors	SEC, FEC, LDA, Congress, Federal Register, USAspending, OpenCorporates, GLEIF, OpenSanctions/OFAC.
4	Entity graph and search	Entity profile, graph edges, pathfinding, related-party discovery, OpenSearch.
5	Stock and investor analysis	Market data abstraction, financial statements, ratios, DCF, comps, technicals, stock report.
6	Claude finance skills	FinanceSkillsGateway, Anthropic adapters, comps/DCF/earnings/coverage workflows, artifact storage.
7	Report editing and review	Editable report workspace, comments, AI enhancement, citation verifier, approvals, exports.
8	P1 data expansion	Regulations.gov, eCFR, RegInfo/OIRA, SAM.gov, IRS 990, CourtListener, FARA.
9	Scale and enterprise hardening	Lakehouse, warehouse, cost dashboards, SSO/SCIM, security reviews, load tests.

13.5 Example Codex tickets

```

Ticket: Build SEC EDGAR connector
Acceptance criteria:
- Uses official SEC APIs and source_contract.yml.
- Supports ticker/CIK lookup, submissions, company facts, filings metadata.
- Stores raw JSON/HTML/XML/PDF before parsing.
- Creates Company, Filing, Security, Person, InsiderTransaction entities/edges.
- Adds rate limiting and retries.
- Adds unit tests with fixtures and integration test marked @integration.

Ticket: Build FinanceSkillsGateway DCF adapter
Acceptance criteria:
- Accepts target entity, financial statement refs, assumptions, scenarios.
- Routes to Anthropic DCF skill when enabled or internal fallback otherwise.
- Produces DCFArtifact with Excel workbook, assumptions JSON, sensitivity table.
- Stores SkillRun, input/output hashes, cost, evidence refs, review status.
- Blocks final report insertion until assumptions and outputs pass validation.

Ticket: Build Report Review Workspace
Acceptance criteria:
- Rich editor with sections, comments, tracked versions, claim cards.
- Enhance selected section with prompt actions.
- Re-run citation verification after edits.
- Reviewer can approve/reject claims and sections.
- Export approved report to PDF, Word, PPT, Excel, and evidence CSV.

```

14. Acceptance Criteria and Launch Gates

14.1 Developer acceptance criteria

- All code passes unit, integration, lint, type, and security checks.
- Every connector has source_contract.yml, fixtures, dry-run mode, rate limits, raw persistence, and evidence refs.
- Every report claim is generated from an EvidenceBundle and can be traced to source records.
- The Review Workspace supports editing, enhancing, commenting, versioning, approving, and re-verifying reports.
- The Claude FinanceSkillsGateway can run at least comps, DCF, earnings analysis, and initiating coverage workflows or internal fallbacks.
- Analyze Stock produces fundamentals, valuation, technicals, catalysts, risks, public-record intelligence, and exportable report.
- Entity profiles show identifiers, graph edges, timeline, evidence, financials, lobbying, contracts, regulations, court/risk, and related parties.
- Cost dashboards show API, warehouse, search, storage, embeddings, and LLM costs by workspace, source, and report.
- Enterprise permissions and tenant isolation pass tests.

14.2 Launch gates

Gate	Requirement
Security	SSO/RBAC/audit logs, secrets management, encryption, tenant isolation, export controls tested.
Data quality	Connector checks, source freshness, schema validation, dedupe, entity-resolution review queue.
Evidence	Every entity/edge/claim has evidence refs or is explicitly labeled assumption/opinion.
Compliance	Restricted source rules for FEC/SAM/licensed data; legal language and investment disclaimers in place.
Financial outputs	Valuation assumptions visible; scenarios included; no guaranteed returns; human approval required.
Cost	Budgets and query guardrails configured; LLM and data-provider spend tracked.
Reliability	Connector retries, dead-letter queues, idempotency, backups, restore test, monitoring, alerting.
User workflow	Search -> analyze -> report -> edit/enhance -> verify -> approve -> export -> watchlist works end to end.

14.3 First enterprise demo acceptance test

- 1 Search for a public company/ticker.
- 2 Resolve entity and show CIK, ticker, LEI/OpenCorporates/GLEIF matches where available.
- 3 Show SEC filings, fundamentals, insider transactions, institutional ownership where available.
- 4 Show lobbying, bills, Federal Register/regulatory exposure, contracts, sanctions/exclusions, and court risk where available.
- 5 Run Analyze Stock in long-term investor mode.
- 6 Run Claude DCF and comps skills or internal fallbacks.
- 7 Generate investor intelligence report with evidence citations.
- 8 Open Review Workspace, edit a section, enhance another section, add a counterargument, refresh citations, and approve claims.
- 9 Export PDF, PowerPoint, Excel model, and evidence appendix.
- 10 Add the entity to a watchlist and create alerts for new filings, contracts, lobbying reports, regulations, and market catalysts.

15. Appendix A - Detailed Backlog

15.1 Epic list

Epic	Stories
Platform foundation	Monorepo, Docker, CI/CD, FastAPI, Next.js, migrations, env config, logging, error tracking.
Enterprise auth	Users, orgs, workspaces, roles, permissions, invites, SSO, SCIM, MFA, API keys.
Source registry	Source model, credentials, contracts, runs, connector status, freshness, rate limits, error dashboard.
Evidence vault	Raw document storage, hashes, EvidenceRef, source records, evidence viewer, retention.
Connectors	SEC, FEC, LDA, FARA, Congress, GovInfo, Federal Register, Regulations.gov, eCFR, RegInfo, USAspending, SAM, IRS 990, CourtListener, sanctions, OpenCorporates, GLEIF, market data.
Ontology and graph	Entities, identifiers, relationships, graph API, pathfinding, graph UI, timeline, evidence-linked edges.
Entity resolution	Deterministic matching, fuzzy/probabilistic matching, review queue, merge/unmerge, conflict detection.
Financial analysis	Market data, statements, ratios, DCF, comps, technical analysis, earnings, insider/institutional ownership, stock report.
Claude skills	Skill registry, managed-agent adapter, skill run logs, DCF/comps/earnings/coverage/diligence/pitch workflows.
Reports	Templates, evidence bundles, claim builder, citation verifier, editor, review workflow, exports.
Alerts	Alert rules, triggers, delivery, watchlists, notification preferences, webhooks.
Portfolio	Portfolio model, CSV upload, positions, risk exposure, catalyst calendar, portfolio report.
Compliance	Restricted data, export approvals, review queues, report language risk, data retention, legal hold.

15.2 Source-specific development requirements

Connector	Must implement
SEC	CIK/ticker lookup; submissions; company facts; filing metadata; fetch filing docs; parse Form 4, 13F, 13D/G, Form D, DEF 14A; rate limiter.
FEC	Candidate/committee search; receipts; disbursements; independent expenditures; bulk loader; contributor restriction flags.
LDA	Registrations; quarterly reports; contribution reports; clients; registrants; lobbyists; issues; agencies; bills; bill linker.
FARA	Registrants; foreign principals; filings; informational materials; receipts/disbursements where available; political activity parser.
Congress	Bills; members; sponsors; cosponsors; committees; actions; amendments; related bills; votes if available.
Federal Register	Documents; proposed/final rules; notices; agencies; CFR refs; RINs; docket IDs; daily watch.
Regulations.gov	Dockets; documents; comments; attachments; submitters; agency IDs; text extraction; org matching.
eCFR	Titles; parts; sections; historical versions; recent changes; regulatory text diffs.
USAspending	Awards; recipients; agencies; subawards; amounts; NAICS/PSC; place of performance; parent recipient data.
SAM.gov	Public opportunities; public entity data; exclusions; assistance listings; sensitive-data guardrails.
IRS 990	XML downloader; parser; nonprofit/foundation entities; officers; grants; grantees; related organizations.
CourtListener	Search; dockets; opinions; parties; attorneys; judges; RECAP metadata and PDFs where available.
Sanctions	OpenSanctions bulk/API; OFAC verification; SAM exclusions; fuzzy screening; risk flags.
OpenCorporates/GLEIF	Company search; profiles; officers; registry IDs; LEIs; parent/child; same-as edges.
Market data	Provider abstraction; prices; OHLCV; corporate actions; fundamentals; transcripts; estimates/options only if licensed.

16. Appendix B - Requirements Checklist for Developers

16.1 Source connector checklist

- source_contract.yml exists and has owner, base URL, auth, rate limits, legal notes, refresh, output entities, quality checks.
- Connector supports dry-run mode and integration mode.
- Connector stores raw source records before parsing.
- Connector uses idempotent upserts and source-specific stable IDs.
- Connector creates EvidenceRefs for all normalized facts.
- Connector logs source run metrics: records fetched, records changed, errors, cost, duration, retries.
- Connector respects rate limits and terms.
- Connector has unit tests with fixtures and at least one integration test marked separately.
- Connector emits data-quality warnings for missing IDs, stale data, parse errors, or schema drift.

16.2 Report workflow checklist

- Report has a template, scope, data cutoff date, evidence bundle, and version.
- Each section has source claims and claim cards.
- Users can edit sections and request enhancement.
- Enhancements preserve or regenerate citations.
- Report changes are versioned and diffable.
- Reviewers can comment, approve, reject, and request changes.
- Citation verifier blocks unsupported facts.
- Final export includes evidence appendix and source list.
- Financial report outputs include assumptions and risks.

16.3 Finance skills checklist

- SkillRegistry includes allowed Claude finance skills and internal fallback skills.
- SkillRun logs input artifacts, output artifacts, model/provider, cost, status, and reviewer.
- DCF/comps/earnings/initiating coverage outputs are stored as structured JSON plus human-readable artifacts.
- Excel workbooks, PowerPoint decks, and PDF/Word reports are versioned and linked to report sections.
- Claude outputs are draft until claim verification and human approval.
- Skill execution is sandboxed and restricted to scoped evidence bundles.
- Users can rerun a skill with changed assumptions and compare versions.

16.4 Data platform checklist

- Raw lake bucket exists with immutable paths and lifecycle policies.
- Parquet lakehouse tables are partitioned and compacted.
- Postgres high-volume tables are partitioned.
- Warehouse queries have budgets and guardrails.
- Embeddings are deduped by chunk hash and model version.

- OpenSearch indexes have shard/retention plans.
- Costs are attributed by tenant, source, connector, report, user, and model.
- Backups and restore tests are implemented.

17. Appendix C - References

- R1.** SEC EDGAR Application Programming Interfaces - <https://www.sec.gov/search-filings/edgar-application-programming-interfaces>
- R2.** OpenFEC API Documentation - <https://api.open.fec.gov/developers/>
- R3.** LDA.gov API Documentation - <https://lda.gov/api/redoc/v1/>
- R4.** USAspending API Endpoints - <https://api.usaspending.gov/docs/endpoints>
- R5.** Anthropic financial-services GitHub repository - <https://github.com/anthropics/financial-services>
- R6.** Anthropic - Agents for financial services - <https://www.anthropic.com/news/finance-agents>
- R7.** Anthropic - Advancing Claude for Financial Services - <https://www.anthropic.com/news/advancing-claude-for-financial-services>
- R8.** Apache Iceberg - <https://iceberg.apache.org/>
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- R11.** Google BigQuery Pricing - <https://cloud.google.com/bigquery/pricing>
- R12.** Snowflake Pricing - <https://www.snowflake.com/en/pricing-options/>
- R13.** Amazon Athena Pricing - <https://aws.amazon.com/athena/pricing/>
- R14.** ClickHouse Cloud Pricing - <https://clickhouse.com/pricing>
- R15.** Dagster - software-defined assets - <https://dagster.io/glossary/software-defined-assets>
- R16.** Prefect Docs - <https://docs.prefect.io/v3/get-started>
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- R18.** Neo4j Knowledge Graph - <https://neo4j.com/use-cases/knowledge-graph/>
- R19.** DuckDB - <https://duckdb.org/>
- R20.** Claude support - Install financial services plugins for Cowork - <https://support.claude.com/en/articles/13851150-install-financial-services-plugins-for-cowork>
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- R23.** OpenAI Codex CLI - <https://github.com/openai/codex>

End of document.